Algorithms for Big Data

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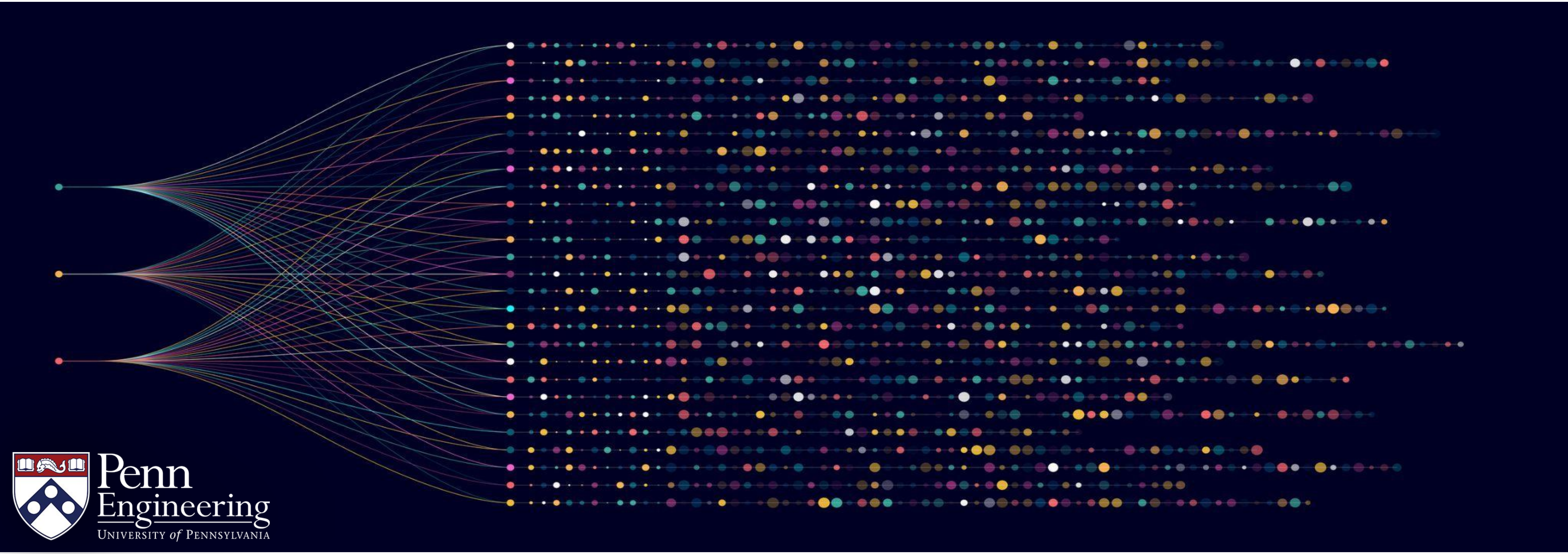
Analysis of the Algorithm



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Objectives: Computing Large Averages

- Analysis of algorithm
- Expectation and Variance analysis

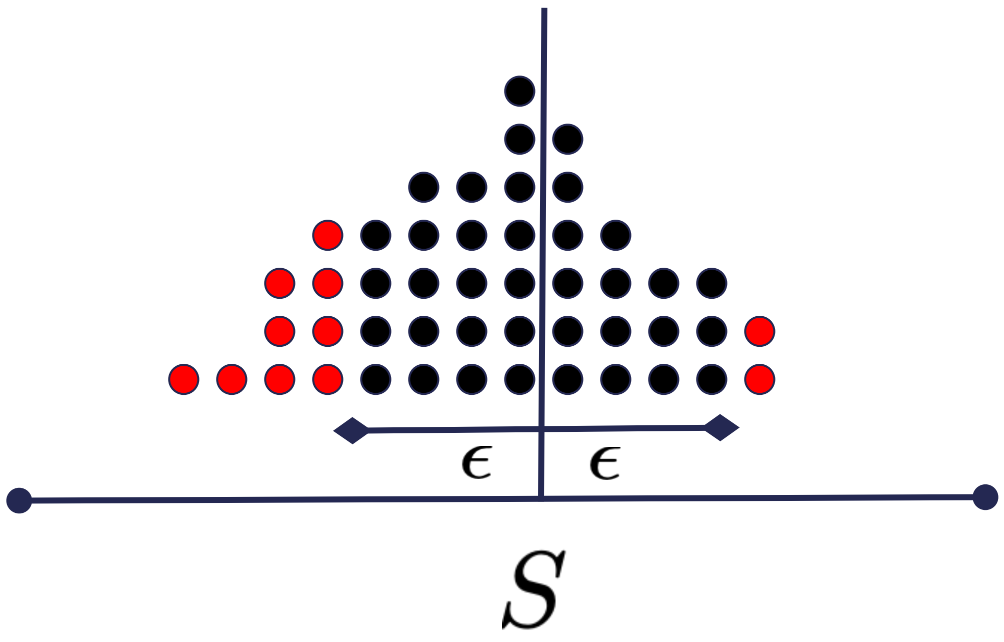
Analysis of the Algorithm

1. Sample q positions:

$$\boldsymbol{i}_1, \ldots, \boldsymbol{i}_q \sim [n]$$

2. Output empirical average:

$$\hat{\boldsymbol{S}} = \frac{1}{q}\sum_{t=1}^{q} X_{\boldsymbol{i}_t}$$



ϵ : accuracy $\quad$ δ: failure probability

Chebyshev's Inequality

- For any random variable $\boldsymbol{Z}$ with expectation $\mathbb{E}[\boldsymbol{Z}] = \mu$

$$\Pr[|\boldsymbol{Z} - \mu| \geq \epsilon] \leq \frac{\mathrm{Var}[\boldsymbol{Z}]}{\epsilon^2}$$

- Exactly what we need:

$$\Pr\left[\left|\hat{\boldsymbol{S}} - S\right| \geq \epsilon\right] \leq \frac{\mathrm{Var}[\hat{\boldsymbol{S}}]}{\epsilon^2}$$

The Plan:

- Compute the expectation of $\hat{\boldsymbol{S}}$
- Compute an upper bound on $\mathrm{Var}[\hat{\boldsymbol{S}}]$
  - Variance will depend on q

$$\Pr\left[\left|\hat{\boldsymbol{S}} - S\right| \geq \epsilon\right] \leq \frac{\mathrm{Var}[\hat{\boldsymbol{S}}]}{\epsilon^2}$$

Analysis via Chebyshev's Inequality

- Simplifying Expectation of $\hat{\boldsymbol{S}}$:

$$\mathop{\mathbb{E}}_{\boldsymbol{i}_1,\ldots,\boldsymbol{i}_q\sim[n]}\left[\hat{\boldsymbol{S}}\right] = \mathop{\mathbb{E}}_{\boldsymbol{i}_1,\ldots,\boldsymbol{i}_q\sim[n]}\left[\frac{1}{q}\sum_{t=1}^{q} X_{\boldsymbol{i}_t}\right]$$

$$= \frac{1}{q}\sum_{t=1}^{q}\mathop{\mathbb{E}}_{\boldsymbol{i}_t\sim[n]}\left[X_{\boldsymbol{i}_t}\right]$$

Analysis via Chebyshev's Inequality

- Simplifying Expectation of $\hat{\boldsymbol{S}}$:

$$\mathop{\mathbb{E}}_{\boldsymbol{i}_1,\ldots,\boldsymbol{i}_q \sim [n]}\left[\hat{\boldsymbol{S}}\right] = \frac{1}{q}\sum_{t=1}^{q} \mathop{\mathbb{E}}_{\boldsymbol{i}_t \sim [n]}\left[X_{\boldsymbol{i}_t}\right]$$

- Expectation of each term:

$$\mathop{\mathbb{E}}_{\boldsymbol{i}_t \sim [n]}\left[X_{\boldsymbol{i}_t}\right] = \sum_{j=1}^{n} \Pr[\boldsymbol{i}_t = j] \cdot X_j = \frac{1}{n}\sum_{j=1}^{n} X_j = S.$$

Chebyshev's Inequality

- For any random variable $\boldsymbol{Z}$ with expectation $\mathbb{E}[\boldsymbol{Z}] = \mu$

$$\Pr[|\boldsymbol{Z} - \mu| \geq \epsilon] \leq \frac{\mathrm{Var}[\boldsymbol{Z}]}{\epsilon^2}$$

- Exactly what we need:

$$\Pr\left[\left|\hat{\boldsymbol{S}} - S\right| \geq \epsilon\right] \leq \frac{\mathrm{Var}[\hat{\boldsymbol{S}}]}{\epsilon^2}$$

Task:

- Compute an upper bound on $\mathrm{Var}[\hat{\boldsymbol{S}}]$ as a function of q
  - As q increases, variance will decrease

- Set q large enough so that

$$\Pr\left[\left|\hat{\boldsymbol{S}} - S\right| \geq \epsilon\right] \leq \frac{\mathrm{Var}[\hat{\boldsymbol{S}}]}{\epsilon^2} \leq \delta.$$

Variance:

- Simplifying Variance:

$$\mathrm{Var}[\hat{\boldsymbol{S}}] = \mathrm{Var}\left[\frac{1}{q}\sum_{t=1}^{q} X_{\boldsymbol{i}_t}\right]$$

$$= \frac{1}{q^2}\sum_{t=1}^{q} \mathrm{Var}\left[X_{\boldsymbol{i}_t}\right]$$

Variance:

- Simplifying Variance:

$$\mathrm{Var}[\hat{\boldsymbol{S}}] = \frac{1}{q^2} \sum_{t=1}^{q} \mathrm{Var}\left[X_{\boldsymbol{i}_t}\right]$$

- Variance of single term:

$$\begin{aligned} \mathrm{Var}\left[X_{\boldsymbol{i}_t}\right] &= \mathbb{E}\left[\left(X_{\boldsymbol{i}_t} - S\right)^2\right] \\ &\leq 1. \end{aligned}$$

Variance:

- Simplifying Variance:

$$\mathrm{Var}[\hat{\boldsymbol{S}}] = \frac{1}{q^2} \sum_{t=1}^{q} \mathrm{Var}\left[X_{\boldsymbol{i}_t}\right] \leq \frac{1}{q}$$

- Variance of single term:

$$\mathrm{Var}\left[X_{\boldsymbol{i}_t}\right] = \mathbb{E}\left[\left(X_{\boldsymbol{i}_t} - S\right)^2\right] \leq 1.$$

Chebyshev's inequality:

- Substituting bound on variance:

$$\Pr\left[\left|\hat{\boldsymbol{S}} - S\right| \geq \epsilon\right] \leq \frac{\mathrm{Var}[\hat{\boldsymbol{S}}]}{\epsilon^2} \leq \frac{1}{q\epsilon^2}.$$

- Set $q = \dfrac{1}{\delta\epsilon^2}$:

$$\Pr\left[\left|\hat{\boldsymbol{S}} - S\right| \geq \epsilon\right] \leq \delta$$

Complete Description of Algorithm:

1. Sample $q = \frac{1}{\delta\epsilon^2}$ positions:

$$\boldsymbol{i}_1, \ldots, \boldsymbol{i}_q \sim [n]$$

2. Output empirical average:

$$\hat{\boldsymbol{S}} = \frac{1}{q}\sum_{t=1}^{q} X_{\boldsymbol{i}_t}$$

- Completely independent of the size of the data!
- Is it optimal?